

★ ANSWER KEY – CONFIDENTIAL ★

UIL COMPUTER SCIENCE – Invitational A 2024

Questions (+6 points for each correct answer, -2 points for each incorrect answer)

1) B	11) A	21) B	31) D
2) D	12) D	22) D	32) A
3) C	13) B	23) A	33) A
4) C	14) C	24) C	34) B
5) A	15) B	25) D	35) B
6) D	16) A	26) D	36) C
7) B	17) B	27) A	37) B
8) E	18) B	28) A	38) E
9) A	19) D	29) D	*39) 32
10) B	20) A	30) E	*40) 14

* See "Explanation" section below for alternate, acceptable answers.

Note: Correct responses are based on **Java SE Development Kit 20 (JDK 20)** from Oracle, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g., "error" is an answer choice) and any necessary Java SE 20 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used.

Explanations:

1.	B	Start converting them all to base 2. You will notice quickly that A and B have different values. After converting C to base 2, you notice that choice B is the "different" choice. Converting the last two to binary proves that assumption.
2.	D	Use order of operations. Perform integer division first, then do modulus, then add the two values. $22 / 7 + 15 \% 8$ $3 + 7 = 10$
3.	C	The five operations will produce the following results: $22 + 8 = 30$ $3 * 2 = 6$ $7 / 3 = 2$ $4 \% 11 = 4$ $12 - 3 = 9$ There are no spaces between values on the same line. A new line will occur after the 6 and the 2. 306 2 49
4.	C	String St = "BACKPACK"; The length of St is 8. So T = 7 We are to print the char #6 in St which is the C
5.	A	true && (true false) Parentheses first - gives us true && true = true
6.	D	Math.sqrt(40) = 6.32455 Math.ceil() rounds it up to the next highest "integer" and returns it as a double 7.0
7.	B	double T = 21 / 2;; T will have the value of 10.0 $7.5 + 8.5 + 10.0 = 26.0$
8.	E	int M = 5; if (M > 10) This if is false { out.print(M); Line not executed M++; Line not executed } if (M < 3) This if is false out.print(M); Line not executed M++; This line IS executed: M now is 6 if (M == 7) This if is false out.print(M); Line not executed out.print(M); 6 is printed
9.	A	Since N is 5, the loop would do this: for(int x = 5; x < 25; x=x+5) out.print(x+1 + " "); While x has values 5 10 15 20 It prints 6 11 16 21
10.	B	int[] food = {5,2,3,4,1,0}; out.print(food[food[0]-food[4]]); food[0]= 5 food[4]= 1 food[5-1] = food[4] = 1
11.	A	M "grabs" the first value (12) from the Scanner String. Then, three values are "passed by." M will be the sum of the original 12 plus 56 plus 2 = 70

12.	D	<pre>for(int i = 1; i <= 30; i = i + 2) if (i%3==0) out.print(i+" ");</pre> The loop goes through all odd numbers 1 through 29. It only prints those odd numbers that are evenly divisible by 3. 3 9 15 21 27
13.	B	<pre>int w = 511; w = 511 w = w<<3; w = 4088 w = w>>2; w = 1022 w = w++ + w++; w = 1022 + 1023 = 2045 out.print(--w); This prints 2044</pre>
14.	C	64 / 8 = 8
15.	B	<pre>numbers.add(96); [96] for(int x=0; x<=100; x=x+12) numbers.add(x); [96,0,12,24,36,48,60,72,84,96] for(int x=1; x<=5; x=x+1) numbers.remove(0); This removes the first 5 elements out.println(numbers); [48,60,72,84,96]</pre>
16.	A	<pre>String yes = "012345678901234"; int P = yes.indexOf("23",5); P = 12 String no = yes.substring(3,5); no = "34" out.println(no + P); This prints 3412</pre>
17.	B	After 0 passes: x=1 y=20 After 1 pass : x=2 y=20 After 2 passes: x=4 y=19 After 3 passes: x=6 y=18 After 4 passes: x=8 y=17 After 5 passes: x=10 y=16 After 6 passes: x=12 y=15 After 7 passes: x=14 y=14 One more is added to x and one more is subtracted from y x=15 y=13 15 * 13 = 195
18.	B	<pre>(12 9) ^ (14 & 10) (1100 OR 1001) XOR (1110 AND 1010) 1101 XOR 1010 0111 = 7</pre>
19.	D	Fibonacci-esque int[]Special = {2,5,7,12,19,31,50,81,131,212,343,... The first two elements are 2 and 5. To get all other elements, add the two elements before that position.
20.	A	<pre>for(int x=0; x<Moe.length; x++) Moe[x] = x*x;</pre> Moe is an array of the squares of the index. 32*32 = 1024
21.	B	<pre>for(int x=0; x<Larry.length; x++) Larry[x] = Moe[x]%5;</pre> Larry is an array of the perfect squares modulus 5 11 * 11 % 5 = 1

22.	D	<pre>for(int x=0; x<Shemp.length; x++) Shemp[x] = Moe[x] - Larry[x];</pre> Shemp is the square of the index minus the square of the index modulus 5 $16*16 - 16*16\%5$ $256 - 1 = 255$
23.	A	<pre>String ten = "JABAAAABBBBZAB"; for(int x=1; x<=100; x++) ten = ten.replaceAll("AB",""); out.print(ten.length());</pre> The first time replaceAll is used, three ABs are replaced. But after that happens, another AB is formed. The loop keeps removing those ABs and continues many more times than needed. The only thing left will be JZ - That is a length of 2
24.	C	Loop iterations x=0 T=6 x=2 T=7 x=4 T=8 x=6 T=9 x=8 T=10 Then x increments by 2, but does not do the loop body. The final value of T is 10
25.	D	<pre>public static int Park(int A, int B) { if (A==B) return (A+B)/3; if (A > B) return Park(A-1, B) + A; return Park(A, B-2) + B; }</pre> Park(10,10) does not recurse ... $(10+10)/3 = 6$
26.	D	Park(7,5) = Park(6,5) + 7 = 9+7= 16 Park(6,5) = Park(5,5) + 6 = 3+6 = 9 Park(5,5) = $(5+5)/3 = 3$
27.	A	Park(3,12) = Park(3,10) + 12 = 32+12= 44 Park(3,10) = Park(3,8) + 10 = 22+10= 32 Park(3,8) = Park(3,6) + 8 = 14+8= 22 Park(3,6) = Park(3,4) + 6 = 8+6= 14 Park(3,4) = Park(3,2) + 4 = 4+4= 8 Park(3,2) = Park(2,2) + 3 = 1+3= 4 Park(2,2) = $(2+2)/3 = 1$
28.	A	<pre>low = [] low.add(60) low=[60] low.add(75) low=[60,75] low.add(30) low=[30,75,60] low.add(12) low=[12,30,60,75] low.add(15) low=[12,15,60,75,30] low.add(90) low=[12,15,45,75,30,90,60] low.add(45) low=[12,15,45,64,30,90,60,75] low.remove() The 12 is removed</pre>
29.	D	After the 12 is removed low=[15,30,45,64,75,90,60]
30.	E	Remove the 15 low=[30,60,45,64,75,90] Remove the 30 low=[45,60,90,64,75]

31.	D	frog = 0 x loop 1: frog loses 1, frog gains 1 x loop 2: frog loses 1, frog gains 2 x loop 3: frog loses 1, frog gains 3 ... x loop 10: frog loses 1, frog gains 10 frog will equal $0+1+2+3+4+5+6+7+8+9 = 45$
32.	A	Instantiating a Mom object sets the Name = "Bella" and the ID=33
33.	A	Instantiating a Son object with two parameters originally sets Name = "Bella" and the ID=33 Name is changed to Dylan, but the 55 is not used. ID is 33
34.	B	Instantiating a Son object with no parameters originally sets Name = "Bella" and the ID=33 The ID is set to 44 after the call to the parent (Mom).
35.	B	U has an integer value in the range [10,15] T has an integer value in the range [3,7] U*T has an integer value in the range [30,105] Z has an integer value in the range [3,10] 7 is the only choice in that range
36.	C	After building the tree: Level 0 contains U Level 1 contains I V Level 2 contains I L Level 3 contains I L N Level 4 contains AN T Level 5 contains A I T Level 6 contains A O Height is 6
37.	B	B starts out = true Every time it goes through the loop, it switches to the opposite boolean value. The loop iterates 38 times. Since this is an even number of iterations, the value will be the original one ... true
38.	E	2 3 0 0 0 0 0 0 0 0 5 6 3 3 3 3 3 3 3 3 8 9 6 6 6 6 6 6 6 6 11 12 9 9 9 9 9 9 9 9 14 15 12 12 12 12 12 12 12 12 17 18 15 15 15 15 15 15 15 15 20 21 18 18 18 18 18 18 18 18 23 24 21 21 21 21 21 21 21 21 26 27 24 24 24 24 24 24 24 24 29 30 27 27 27 27 27 27 27 27 $17+18+15+15+15 = 8$

39.	32	Here is the evolution of the stack: Bottom -> Top <pre> push 10 [10, 8] push 8 [10, 8] push 12 [10, 8, 12] pop [10, 8] push 11 [10, 8, 11] push 6 [10, 8, 11, 6] push 7 [10, 8, 11, 6, 7] pop [10, 8, 11, 6] pop [10, 8, 11] push 3 [10, 8, 11, 3] push 5 [10, 8, 11, 3, 5] pop [10, 8, 11, 3] 10+8+11+3 = 32 </pre>
40.	14	<pre> + 8 * 2 / 60 + - 7 2 * 3 5 + 8 * 2 / 60 + - 7 2 15 + 8 * 2 / 60 + 5 15 + 8 * 2 / 60 20 + 8 * 2 3 + 8 6 14 </pre>